

SOCIOL 723: Social Statistics II

Week 11, Synthetic Control and Its Extensions

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Today

The Synthetic Control Estimator

Extensions

Application: Prop 47 and Crime

Practice

★ = essential, you must understand this

★ = for understanding, not examined

When There Is Only One Treated Unit ★

Difference-in-differences needs a control *group*. What if the treated unit is California, or East Germany, or a single firm?

- Comparative case studies have always done this informally: “compare California to the rest of the country.” The choice of comparison was made by the author, and the reader had to trust it.
- **Synthetic control** makes that choice a data-driven, transparent, pre-registered procedure: construct a weighted average of untreated units that reproduces the treated unit’s pre-treatment trajectory.
- “Arguably the most important innovation in the policy evaluation literature in the last 15 years” (Athey and Imbens 2017).

The identifying idea is not new. What is new is that the weights are chosen by an algorithm, reported in a table, and checked against placebo units.

Plan

Goal: a counterfactual for one treated unit, how to judge it, and what to do when the classic method does not fit.

1. The estimator, its weights, and why the constraints matter; Proposition 99 throughout, rebuilt from the data.
2. What justifies it: a factor model, where bias comes from, and why the length of the pre-period does the work (a simulation).
3. Inference with one treated unit: placebos, in-time placebos, leave-one-out, conformal tests, prediction intervals.
4. Extensions: penalized and augmented SC, synthetic DiD, matrix completion and interactive fixed effects, many treated units.
5. Applications: German reunification; Prop 47 and crime.

Data: `tidysynth::smoking`, 39 states 1970–2000; scripts `figs/w11_prop99.R`, `figs/w11_sims.R`.

Synthetic Control

Setup ★

Unit 1 is treated from period $T_0 + 1$ onward. Units $2, \dots, J + 1$, the **donor pool**, are never treated. The estimand is

$$\tau_{1t} = Y_{1t}(1) - Y_{1t}(0), \quad t > T_0.$$

We observe $Y_{1t}(1)$; the whole problem is imputing $Y_{1t}(0)$.

The synthetic control

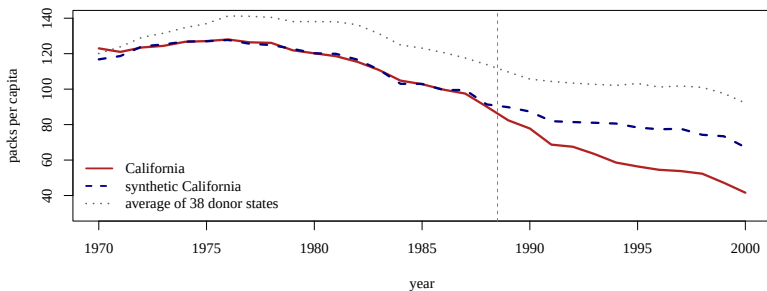
Choose weights $w_2, \dots, w_{J+1}$ with $w_j \geq 0$ and $\sum_j w_j = 1$ so that the weighted donors match the treated unit *before* treatment:

$$\sum_{j \geq 2} w_j Y_{jt} \approx Y_{1t} \quad \text{for } t \leq T_0, \quad \sum_{j \geq 2} w_j X_j \approx X_1,$$

then estimate $\hat{\tau}_{1t} = Y_{1t} - \sum_{j \geq 2} w_j Y_{jt}$ for $t > T_0$.

Running Example: Proposition 99

California raised its cigarette tax by 25 cents a pack in 1989 and funded anti-smoking campaigns. Outcome: packs sold per capita. Donors: 38 states with no large tobacco program or tax increase in the period.



- The donor average is a poor counterfactual: California was already falling faster before 1989 (1988: 90.1 against 113.8).
- The synthetic California tracks it to 1988 and then separates (figs/w11_prop99.R).

The Optimization Problem, Explicitly ★

Let $\mathbf{X}_1$ be a $k \times 1$ vector of pre-treatment predictors for the treated unit and $\mathbf{X}_0$ the $k \times J$ matrix for the donors. Choose weights

$$\mathbf{W}^*(\mathbf{V}) = \arg \min_{\mathbf{W} \in \mathcal{W}} (\mathbf{X}_1 - \mathbf{X}_0 \mathbf{W})' \mathbf{V} (\mathbf{X}_1 - \mathbf{X}_0 \mathbf{W}), \quad \mathcal{W} = \left\{ \mathbf{W} \geq 0, \sum_j w_j = 1 \right\}.$$

- $\mathbf{V}$ is a diagonal matrix of **predictor importance** weights. It is itself chosen, usually to minimize pre-treatment prediction error of the *outcome*:

$$\mathbf{V}^* = \arg \min_{\mathbf{V}} \sum_{t \leq T_0} (Y_{1t} - \sum_j w_j^*(\mathbf{V}) Y_{jt})^2.$$
- So there are two nested optimizations, and the inner one is a constrained quadratic program. The constraint set $\mathcal{W}$ is a simplex.
- **This nesting is where researcher degrees of freedom hide.** Which predictors enter $\mathbf{X}$, and how many pre-periods are used, can move the answer. Pre-register the choice or report the full space.

The Weights and the Balance on Prop 99

Predictors as in Abadie, Diamond, and Hainmueller (2010): income, price, share aged 15–24 (1980–88), beer (1984–88), sales in 1975, 1980, 1988.

state	ours	ADH
Utah	0.342	0.334
Nevada	0.238	0.234
Montana	0.209	0.199
Colorado	0.149	0.164
Connecticut	0.062	0.069
other 33	0	0

predictor	CA	synth.	donors
log income	10.08	9.85	9.83
retail price	89.4	89.4	87.3
share 15–24	.174	.174	.173
beer	24.3	24.2	23.7
sales 1975	127.1	127.0	136.9
sales 1980	120.2	120.2	138.1
sales 1988	90.1	91.4	113.8

- Five states carry all the weight. Pre-period RMSPE 1.78 packs.
- Average gap 1989–2000: –18.8 packs (about –24%); in 2000: –25.6.
- Income is not matched: California is richer than any combination.

The Factor-Model Justification ★

Matching California's *past* is only useful if it tells us something about California's missing *future*. Why should it? The factor model is the answer: it states the conditions under which a good pre-treatment fit licenses that extrapolation.

Suppose untreated outcomes follow a linear factor model

$$Y_{jt}(0) = \delta_t + \boldsymbol{\theta}'_t \mathbf{Z}_j + \boldsymbol{\lambda}'_t \boldsymbol{\mu}_j + \epsilon_{jt},$$

with $\boldsymbol{\lambda}_t$ common (unobserved) factors and $\boldsymbol{\mu}_j$ unit-specific loadings.

- **DID** allows arbitrary common time shocks δ_t , but requires that the remaining term $\boldsymbol{\lambda}'_t \boldsymbol{\mu}_j$ not generate *differential* trends between treated and control units. Synthetic control weakens that by matching on the loadings instead.

The Factor-Model Justification ★ (cont.)

- Abadie, Diamond, and Hainmueller (2010) show that if a convex combination of donors reproduces the treated unit's outcomes over a *long* pre-period *and* its observed predictors, then under regularity conditions, enough pre-periods relative to the noise, and a factor structure of bounded rank, the weighted donors also approximately match the **unobserved loadings** μ_1 , and a bound on the post-treatment bias shrinks with the number of pre-periods.
- **Length of the pre-period is doing the identification work.** A two-period match proves nothing; a twenty-period match is informative. This is the single most important design consideration.

Where the Bias Comes From

Write the factor model for treated and synthetic units after treatment ($t > T_0$):

$$Y_{1t}(0) - \sum_j w_j Y_{jt}(0) = \boldsymbol{\theta}'_t(\mathbf{Z}_1 - \sum_j w_j \mathbf{Z}_j) + \boldsymbol{\lambda}'_t(\boldsymbol{\mu}_1 - \sum_j w_j \boldsymbol{\mu}_j) + (\epsilon_{1t} - \sum_j w_j \epsilon_{jt}) \quad \delta_t \text{ cancels: } \sum_j w_j = 1$$

- A good pre-period fit matches $\mathbf{Z}$ and, over many periods, can only be achieved by matching the loadings $\boldsymbol{\mu}_1$: the middle term is small.
- With few pre-periods, the weights can instead **fit the noise** ϵ_{1t} : a tight fit that says nothing about $\boldsymbol{\mu}$.
- The bound in Abadie, Diamond, and Hainmueller (2010) falls with T_0 only if the fit is good, and rises with the number of donors J : more donors make it easier to fit noise (Abadie 2021).
- DiD is the case $w_j = 1/J$ plus a level shift: unbiased only if $\boldsymbol{\mu}_1$ equals the donors' average loading.

Simulation: the Pre-Period Does the Work

Two factors, 20 donors, the treated unit's loadings inside the donors' hull but far from their average; true effect 0; 1,000 draws (`figs/w11_sims.R`).

pre-periods T_0	3	5	10	20	40
<i>low noise</i> ($\sigma = 0.25$): root mean squared error					
synthetic control	0.44	0.38	0.27	0.19	0.14
difference-in-differences	0.39	0.40	0.39	0.43	0.49
<i>high noise</i> ($\sigma = 1$)					
synthetic control	0.68	0.67	0.66	0.56	0.47
difference-in-differences	0.78	0.67	0.60	0.55	0.56
SC pre-period RMSPE	0.22	0.38	0.63	0.79	0.90

- With little noise, SC improves steadily with T_0 ; DiD's bias does not shrink.
- With much noise and short T_0 , SC is no better than DiD, and its pre-period fit (0.22) is **tighter than the noise** ($\sigma = 1$): it fit the noise. A suspiciously perfect fit is a warning, not a reassurance.

Why the Constraints Matter ★

The restrictions $w_j \geq 0$ and $\sum_j w_j = 1$ look like technicalities. They are the heart of the method.

- **No extrapolation:** the synthetic unit is a *convex combination* of real units, so it lies inside the donor pool's support. Unconstrained regression would happily assign weights of +3 and -2 and predict outside anything ever observed.
- **Sparsity:** the constraints usually produce weights on a handful of donors and zero on the rest. You can print the weight table and let readers judge whether the comparison is sensible.
- **Transparency:** the counterfactual is a named, weighted set of real places, not a regression coefficient.
- If no convex combination matches the treated unit, if California is richer than every donor, **the method tells you**, through poor pre-treatment fit. That is a feature.

What the Constraints Prevent, on Prop 99

Drop $w_j \geq 0$ and keep $\sum_j w_j = 1$: with 38 donors and 19 pre-years, infinitely many weightings fit the pre-period exactly. Take the smallest one (minimum norm)

(figs/w11_prop99.R):

	SC weights	regression weights
nonzero donors	5	38
negative weights	0	17 (as low as -0.13 , Tennessee)
pre-period fit	RMSPE 1.78	exact
average gap 1989–2000	-18.8	-15.5

- An exact fit with $J > T_0$ is guaranteed and therefore uninformative: it fits the noise.
- Negative weights mean the counterfactual California is “West Virginia plus Connecticut minus Tennessee minus Mississippi”: extrapolation no reader can inspect (Abadie 2021).

Relation to Difference-in-Differences ★

DID is synthetic control with the weights fixed at $1/J$ (or population shares), and with an added intercept shift.

- DID assumes parallel trends between the treated unit and a *pre-specified* average of controls.
- Synthetic control *searches* for a weighting under which pre-treatment trajectories coincide, and then assumes that coincidence continues. The factor model on the previous frame states when that assumption is warranted.
- So the two methods sit on a spectrum: DID trusts a pre-specified comparison and an intercept shift; synthetic control trusts a fitted comparison and no intercept. Synthetic DID (later today) combines them.

Inference With One Treated Unit ★

Conventional standard errors are meaningless here: $n_{\text{treated}} = 1$, and the usual asymptotics run in the number of treated units. Instead, **permutation inference**.

Placebo (in-space) tests

Pretend each donor was the treated unit and estimate its “effect.” If the real treated unit’s post-treatment gap is extreme relative to the placebo distribution, that is evidence.

For each unit j compute

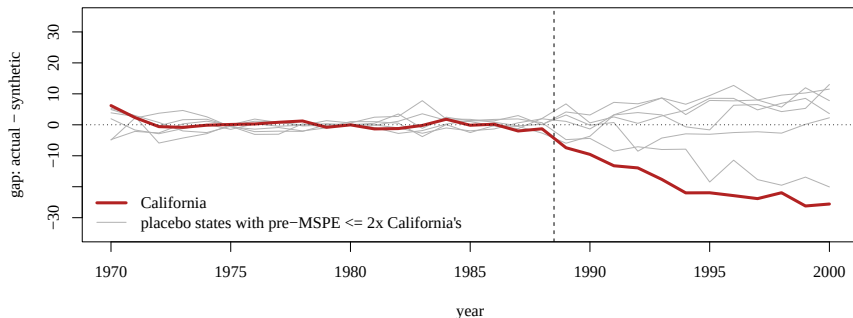
$$\text{RMSPE}_j^{\text{pre}} = \sqrt{\frac{1}{T_0} \sum_{t \leq T_0} (Y_{jt} - \hat{Y}_{jt}^N)^2}, \quad \text{RMSPE}_j^{\text{post}} = \sqrt{\frac{1}{T-T_0} \sum_{t > T_0} (Y_{jt} - \hat{Y}_{jt}^N)^2},$$

and rank units by the **ratio** $\text{RMSPE}_j^{\text{post}}/\text{RMSPE}_j^{\text{pre}}$.

Inference With One Treated Unit ★ (cont.)

- The ratio, rather than the raw post-treatment gap, is what makes placebos comparable: a donor the method could not fit will show a large gap for reasons unrelated to treatment. **Discard poorly fitting placebos.**
- The rank of the treated unit among the $J + 1$ units gives a permutation-style p -value: with 38 donors the smallest attainable value is $1/39 \approx 0.026$. **Call it a placebo rank, not an exact p -value:** exactness would require treatment to have been randomly assigned among these units under a sharp null, which it was not.
- **In-time placebo:** re-run pretending treatment began earlier. A gap should not appear before the real intervention.
- **Leave-one-donor-out:** drop each positively weighted donor in turn. If the result rests on one unit, say so.

Placebos on Prop 99



- Pre-period MSPE: California 3.17; median donor 15.4. Only 6 donors fit as well as twice California's MSPE (shown).
- Post/pre MSPE ratio: California 124, largest of 39 (next: Georgia, 47). Placebo rank $1/39 = 0.026$, the smallest attainable.

In-Time Placebo and Leave-One-Out on Prop 99

Outcome-only weights (all 19 pre-years) give nearly the same answer: average gap -19.5 , pre-RMSPE 1.66 (figs/w11_prop99.R).

In-time placebo: pretend the law came in 1980, fit on 1970–79.

- Fit 1970–79: RMSPE 0.84. Average gap 1980–88: -3.4 packs.
- Not zero: California was already pulling away in the 1980s, by about a sixth of the post-1989 effect. Report it.

Leave-one-out: drop each donor with positive weight, refit.

dropped	Utah	Montana	Nevada	Connecticut	N. Hampshire	Colorado
average gap	-19.1	-18.0	-19.2	-20.5	-19.8	-19.6

No single donor drives the estimate.

Conformal Inference (Chernozhukov, Wüthrich, and Zhu 2021)

Placebos permute *units*. Conformal inference permutes *time*, for one hypothesis at a time.

1. Hypothesis H_0 : $\tau_t = \tau_t^0$ for $t > T_0$ (start with $\tau^0 = 0$). Under H_0 , $Y_{1t} - \tau_t^0$ is the untreated outcome in every period.
2. Fit the weights on *all* T periods using that series; residuals $\hat{u}_t$.
3. Statistic $S = \text{mean } |\hat{u}_t|$ over post-periods. Recompute it for every cyclic shift of the residuals in time; the p -value is the share at least as large.
4. Invert: the set of τ^0 not rejected is a confidence set.
 - Valid when residuals are exchangeable (or weakly dependent) over time.
 - Prop 99, H_0 : no effect in any year: $S = 13.5$, $p = 0.097$ with 31 shifts. Weaker evidence than the placebo rank: here the null is tested against the state's own residual noise.

Prediction intervals that combine weight-estimation and post-period uncertainty: Cattaneo, Feng, and Titiunik (2021), `scpi`.

The Canonical Application: California Proposition 99

Abadie, Diamond, and Hainmueller (2010) study California's 1988 tobacco control programme.

- **Outcome:** per-capita cigarette sales, 1970–2000. **Donor pool:** the 38 states that never adopted comparable programmes over the period, states with their own large tax increases or programmes are excluded *a priori*, on substantive grounds.
- **Synthetic California** is a weighted average of Colorado, Connecticut, Montana, Nevada, and Utah. The weight table is printed in the paper, and readers can judge whether that is a sensible California.
- Pre-1988 trajectories are nearly indistinguishable; afterwards they diverge sharply, implying roughly 26 fewer packs per capita per year by 2000.

The Canonical Application: California Proposition 99 (cont.)

- In the placebo distribution, California's post/pre RMSPE ratio is the largest of all 39 units, giving $p = 1/39$.

Notice how much of the credibility comes from things a reader can check by eye: the weight table, the pre-treatment overlap, and the placebo plot.

In the Literature: German Reunification

Abadie, Diamond, and Hainmueller (2015)

The question: what did reunification in 1990 cost West Germany? GDP per capita (PPP, 2002 USD); 16 OECD donors; predictors trained on 1971–80, validated on 1981–90.

donor	weight
Austria	0.42
United States	0.22
Japan	0.16
Switzerland	0.11
Netherlands	0.09

- About $-1,600$ USD per person per year, 1990–2003; 2003 GDP per capita 12% below synthetic.
- In-time placebo, reunification in 1975: no gap.
- Post/pre RMSPE ratio about 16, largest of 17: $p = 1/17 \approx 0.06$.
- Leave-one-out: dropping the US gives the smallest effect, about -630 USD per year.

Regression weights: Greece -0.09 , Portugal -0.08 , Italy -0.05 .

Replications across software packages and variable orderings do not reproduce the published weights (Klößner et al. 2018): $\mathbf{V}$ is poorly identified.

Diagnostics and Failure Modes ★

- **Poor pre-treatment fit** invalidates the design. Report the pre-treatment RMSPE and plot the two trajectories. If they do not overlap before treatment, stop.
- **Interpolation bias:** donors very unlike the treated unit can be combined to match on average while differing on everything that matters. Restrict the donor pool on substantive grounds *before* estimating.
- **Overfitting the pre-period:** matching noise rather than structure. A long pre-period and few donors is the safe configuration.
- **Contamination:** donors affected by the treatment (spillovers) or by their own interventions must be excluded.
- **Specification search:** the choice of predictors and pre-periods can move results. Pre-register it, or report the full space.

Covariates or Lagged Outcomes?

- If *every* pre-period outcome enters $\mathbf{X}_1$ separately, the optimal $\mathbf{V}$ puts all its weight on them and the covariates become irrelevant, however predictive (Kaul, Klößner, Pfeifer, and Schieler 2022).
- Prop 99 illustrates both choices: ADH predictors give -18.8 ; all 19 pre-period outcomes alone give -19.5 .
- Matching on covariates protects against fitting noise in the outcome path; matching on outcomes protects against unmeasured loadings. Report both when they differ.
- $\mathbf{V}$ is chosen by a nested, non-convex optimization; different software returns different $\mathbf{V}$ and weights for the same data (Klößner et al. 2018). Report the weights, not only the gap.

The Basque Country: Two Good Fits, Two Answers

Abadie and Gardeazabal (2003): GDP per capita in the Basque Country after terrorism began around 1970; 16 other Spanish regions as donors (`Synth::basque; figs/w11_more.R`).

	predictors (Synth example)	pre-period outcomes only
weights	Cataluña 0.851, Madrid 0.149	Madrid 0.483, Balears 0.311, Rioja 0.206
pre-period RMSPE	0.094	0.076
gap 1975–97 (thousand 1986 USD)	−0.70	−1.04 (−11.5%)

- Both fit the pre-period almost perfectly, with *disjoint* donors except Madrid. The weights are not unique when several combinations track the treated unit (Abadie 2021).
- The sign is robust; the size depends on the specification. Report the specifications you tried.

Extensions

When the Convex Hull Is Empty

- **Augmented synthetic control** (Ben-Michael, Feller, and Rothstein 2021): when no convex combination fits, allow a *bias correction*, fit an outcome model to the residual imbalance, exactly the augmentation logic of AIPW in Week 6. Relaxes the no-extrapolation constraint by a controlled amount.
- **Ridge-augmented** versions penalize how far the weights stray from uniform, trading a little extrapolation for much better fit.
- **Generalized synthetic control** (Xu 2017): estimate an interactive fixed effects model $Y_{it}(0) = \lambda'_t \mu_i + \dots$ on the untreated units and use it to impute $Y_{it}(0)$ for the treated. Handles *multiple* treated units and staggered timing, and gives conditional standard errors via a parametric bootstrap.
- **Synthetic DID** (Arkhangelsky et al. 2021): unit weights *and* time weights, with a DID-style intercept. Doubly robust in spirit and often the best default now.

Penalized Synthetic Control (Abadie and L'Hour 2021)

SC can match a unit with a combination of donors that are each far from it. Penalize distance to each donor:

$$\min_{\mathbf{W}} \left\| \mathbf{X}_1 - \sum_j W_j \mathbf{X}_j \right\|^2 + \lambda \sum_j W_j \left\| \mathbf{X}_1 - \mathbf{X}_j \right\|^2, \quad W_j \geq 0, \sum_j W_j = 1.$$

- $\lambda \rightarrow 0$: synthetic control. $\lambda \rightarrow \infty$: nearest-neighbour matching (Week 6). In between: a combination of *nearby* donors.
- With $\lambda > 0$ the solution is unique and uses at most $p + 1$ donors.
- Built for many treated units each with its own synthetic control (disaggregated data: schools, counties, individuals).

Augmented Synthetic Control, Written Out (Ben-Michael, Feller, and Rothstein 2021)

Fit an outcome model $\hat{m}(\cdot)$ on the donors and correct SC by the model's estimate of what the imperfect fit leaves:

$$\begin{aligned}\hat{Y}_{1T}^{\text{aug}}(0) &= \underbrace{\sum_j \hat{w}_j Y_{jT}}_{\text{synthetic control}} + \underbrace{\left(\hat{m}_{1T} - \sum_j \hat{w}_j \hat{m}_{jT} \right)}_{\text{predicted cost of the imbalance}} && \text{bias correction, as in Week 6} \\ &= \hat{m}_{1T} + \sum_j \hat{w}_j (Y_{jT} - \hat{m}_{jT}) && \text{same thing, doubly robust form}\end{aligned}$$

- With a ridge outcome model on the pre-period outcomes, the result is a synthetic control whose weights may go slightly negative: a controlled amount of extrapolation, set by the ridge penalty.
- Perfect pre-period fit: the correction is zero. Poor fit: the model steps in.
- Prop 99 (`augsynth`, ridge): average effect -16.0 ; remaining pre-period imbalance 4.6% of what equal weights leave.

Synthetic Difference-in-Differences (Arkhangelsky et al. 2021)

Weight units *and* periods, then run a weighted two-way regression:

$$(\hat{\tau}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg \min \sum_i \sum_t (Y_{it} - \mu - \alpha_i - \beta_t - D_{it}\tau)^2 \hat{\omega}_i \hat{\lambda}_t.$$

- $\hat{\omega}$: SC-type unit weights, but with an intercept (levels need not match) and a ridge penalty (spread the weight).
- $\hat{\lambda}$: period weights that make the pre-period resemble the post-period for the controls. DiD: $\omega = 1/J$, $\lambda = 1/T_0$. SC: no α_i , λ on the post-period only.

Prop 99, synthdid	DiD	SC	SDiD
estimate (packs per capita)	-27.3	-19.6	-15.6

Matches their Table 1. SDiD's time weights fall on 1986–88 (0.37, 0.21, 0.43): the years most like the post-period. Unit weights spread over more states (Nevada 0.12, New Hampshire 0.11, ...).

One Objective, Many Estimators

Athey, Bayati, Doudchenko, Imbens, and Khosravi (2021): write the untreated outcomes as a matrix with the treated cells missing, and impute them.

method	how $Y(0)$ is imputed	software
DiD / TWFE	$\alpha_i + \lambda_t$	fixest
synthetic control	weighted donors, weights from the pre-period	tidysynth
interactive fixed effects	$\alpha_i + \lambda_t + \mathbf{\lambda}'_t \boldsymbol{\mu}_i$, r factors by cross-validation	gsynth (Xu 2017)
matrix completion	low-rank $\mathbf{L}$, nuclear-norm penalty	fect

- They differ in how they regularize, not in what they estimate.
- Factor-model imputation handles many treated units and staggered timing, and nests TWFE ($r = 0$): Week 10's imputation estimator with factors added (Liu, Wang, and Xu 2024).
- Many treated units, one SC each: average imbalance and unit imbalance trade off; Ben-Michael, Feller, and Rothstein (2022) weight both (partially pooled SC).

Interactive Fixed Effects, Written Out (Xu 2017)

Untreated outcomes follow $Y_{it}(0) = \mathbf{X}'_{it}\boldsymbol{\beta} + \alpha_i + \xi_t + \boldsymbol{\lambda}'_t\boldsymbol{\mu}_i + \epsilon_{it}$, with r factors.

1. Fit the model on **control units only**, all periods: $\hat{\boldsymbol{\beta}}$, $\hat{\boldsymbol{\lambda}}_t$.
2. For each treated unit, estimate its loadings $\hat{\boldsymbol{\mu}}_i$ from its **pre-treatment** periods.
3. Impute $\hat{Y}_{it}(0)$ after treatment; $\hat{\tau}_{it} = Y_{it} - \hat{Y}_{it}(0)$; average.
4. Choose r by cross-validation on held-out pre-period cells. $r = 0$ is two-way fixed effects imputation (Week 10).

Election-day registration and turnout (47 states, 1920–2012; 9 adopt EDR at different times) (`gsynth::turnout`; `figs/w11_more.R`):

estimator	effect on turnout (pts)
TWFE	0.78 (3.18)
imputation, $r = 0$	0.83 (5.87)
interactive fixed effects, $r = 2$ (cross-validated)	4.90 (2.18)

Adopting states (Maine, Minnesota, Wisconsin, ...) were on different long-run turnout paths; two factors capture them.

Many Treated Units, Staggered Timing (Ben-Michael, Feller, and Rothstein 2022)

Treated units $i = 1, \dots, J_1$, each with SC weights γ_i and its own pre-period imbalance $\mathbf{d}_i(\gamma_i)$. The error of the *average* effect depends on

$$\underbrace{\left\| \frac{1}{J_1} \sum_i \mathbf{d}_i(\gamma_i) \right\|^2}_{\text{pooled imbalance}} \quad \text{and} \quad \underbrace{\frac{1}{J_1} \sum_i \|\mathbf{d}_i(\gamma_i)\|^2}_{\text{unit imbalance}}.$$

- Separate SCs minimize the second only; one SC for the average minimizes the first only. **Partially pooled** SC minimizes $\nu \cdot \text{pooled} + (1 - \nu) \cdot \text{unit}$.
- Adds unit fixed effects (intercept shifts) as an option: SC on demeaned outcomes, the Ferman–Pinto fix.
- Teacher collective bargaining laws: minimal effects on school spending.
`augsynth::multisynth`.

Spillovers, Anticipation, and the Donor Pool

SC assumes the donors are untouched by the treatment and the treated unit did not react early (Abadie 2021).

- **Spillovers:** smokers buying cigarettes across the border (Nevada received 0.24 of the weight for California). If Nevada's sales rose because of Prop 99, synthetic California is too high, and the effect is overstated. State the direction of the bias.
- **Remedy:** drop neighbours from the pool and compare; in Harknett et al. (Week 10) workers just outside Seattle showed smaller effects, consistent with spillover.
- **Anticipation:** if behaviour changed when the measure was debated, backdate the intervention to the announcement.
- **Other shocks to the treated unit** at the same time (Realignment before Prop 47) are part of what SC estimates.

Recent Directions

- **Imperfect fit:** with an imperfect pre-period fit SC stays biased as T_0 grows if treatment relates to the loadings; demeaning first (SC on deviations from unit means) can beat DiD (Ferman and Pinto 2021).
- **Many controls:** with many periods and many donors, weights spread thinly can still recover the loadings (Ferman 2021).
- **Design-based:** if treatment were randomly assigned, standard SC is biased; a modified estimator is unbiased with an exact variance (Bottmer, Imbens, Spiess, and Warnick 2024).
- **Choosing where to treat:** when you can pick the treated units (a pilot in a few cities), choose them by synthetic control, not by coin flip (Abadie and Zhao 2026).
- **Proximal SC:** donors not in the synthetic unit serve as proxies for the latent factors (Shi et al. 2026).

Prop 47

In the Literature: Bartos and Kubrin (2018)

The question. California's Proposition 47 (November 2014) reclassified some drug and property felonies as misdemeanors, shrinking prison and jail populations. Did crime rise? Opponents pointed to a 2015 uptick.

The data. FBI Uniform Crime Reports, state rates per capita, 1970–2015: seven offenses, each with its own synthetic California. 44 pre-period years; one post-period year (2015).

The design. Donor pool: the other 49 states, since none adopted a comparable reform. Weights from pre-period crime trends. Effect: the 2015 gap.

Decision rule. A gap smaller than the pre-period RMSPE is treated as no effect; placebo ranks among the 50 states give the p -value ($5/50 = 0.10$).

Bartos and Kubrin: Results

offense	2015 gap	placebo rank
homicide, rape, aggravated assault, robbery	within pre-period RMSPE	—
burglary	within pre-period RMSPE	—
larceny	under 10% above synthetic	4 of 50, $p = 0.08$
motor vehicle theft	about 20% above synthetic	13 of 50, $p = 0.26$

- Violent crime: no detectable effect. Larceny: a modest increase, weakly distinguished from placebo variation.
- Leave-out check: the larceny gap disappears when the heaviest donors are removed.

Bartos and Kubrin: Reading It Critically

- **One post-period year.** The estimate is a single gap, compared with 44 years of fit; effects that build later are invisible, and one year's shock (2015 was a high-crime year in many places) is hard to separate.
- **Read the weight table.** The text names New York, Michigan, Nevada, and New Jersey as larceny's synthetic California; the appendix gives Nevada 0.479, New York 0.406, Colorado 0.095, Massachusetts 0.019, and zero for Michigan and New Jersey. The weights are the counterfactual: check them.
- **Compound treatment.** 2011 Realignment moved prisoners to county jails three years earlier; the authors skip in-time placebos because Realignment would contaminate them.
- **Several outcomes, several tests:** seven offenses, one marginal rejection at 10%.

Same template later used for California's sanctuary law (Kubrin and Bartos 2020).

Practice

Choosing Among Them

Situation	Method	R
One treated unit, good pre-fit	classic synthetic control	<code>Synth</code> , <code>tidysynth</code>
One treated unit, poor pre-fit	augmented / ridge SC	<code>augsynth</code>
Several treated units, staggered	generalized SC	<code>gsynth</code>
Many units, want DID robustness	synthetic DID	<code>synthdid</code>
Many units, staggered, covariates	Callaway–Sant’Anna (Wk 10)	<code>did</code>

- These are not rivals so much as points on a spectrum between “trust the weights” and “trust the outcome model.” The augmented versions deliberately use both, the same insurance logic as AIPW.
- Report the classic estimator alongside whichever you prefer, so readers can see how much the extrapolation is doing.

Reporting a Synthetic Control Study ★

1. The donor pool, and why each unit is in it or excluded.
2. The predictors used for matching and the pre-treatment window.
3. **The weight table.** Which units, with what weights.
4. A balance table: treated versus synthetic on every predictor.
5. The trajectory plot, treated versus synthetic, with the intervention marked.
6. The gap plot, with placebo gaps overlaid.
7. Pre- and post-treatment RMSPE, their ratio, and the permutation p -value.
8. In-time placebo and leave-one-donor-out robustness.

The trajectory plot is the result. If the pre-treatment lines do not sit on top of one another, no test rescues the design.

Looking Ahead

- Synthetic control completes the design-based half of the course. Weeks 5–11 have all been variations on one question: *where does the counterfactual come from?*
- Notice the recurring structure: choose weights on comparison units (Week 6), choose an exogenous slice of variation (Weeks 7, 9), choose a differencing scheme (Week 10), choose a weighted donor pool (Week 11).
- **Weeks 12–13** change the question again: what happens when prediction rather than identification is the goal, and what, if anything, flexible prediction can contribute back to causal work.

Looking ahead

Weeks 12 and 13 turn to machine learning: first as a prediction tool, then as machinery for estimating causal effects without committing to a functional form.

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